

Quantifying Client Dependence in a Federated Water Distribution Network — v2

Interventional sensitivity, hydraulic unmixing, finite-difference calibration, and the jval_ validation suite*

FedWater handoff note | supersedes v1 | Graeme network, 113 junctions / 164 pipes / 5 districts

1. Where this stands

Result. Cross-district dependence is measured by intervention, not inferred from correlation. Inverting an interventionally-measured district Jacobian recovers true district-demand similarity at rank correlation 0.939 (MAE 0.052) against 0.007–0.013 for every model-free alternative tried — a gain of +0.767 rank correlation and 15x on MAE. The result is invariant to the finite-difference step size over 2.3 decades, and the step size is now chosen by a physics-derived criterion rather than by hand.

Why it matters. Correlation is not preserved under linear mixing. This is the single cause behind a long sequence of negative results (sector tests, drift attribution, Granger direction): no choice of *which mixed signal to observe* could have worked, because the mixing had to be inverted rather than avoided.

What changed since v1. Every headline number in v1 was recomputed on an explicitly pinned world set. Four v1 claims do not survive and are corrected in Section 6. The method does; the recommendation does.

Quantity	v2 value	Evidence
Recovery of true demand similarity	rank corr 0.939 , MAE 0.052	40 worlds, pinned set, variance placement, delta = 0.5 x mean_d
Gain over best Jacobian-free method	+0.767 rank corr; 15x MAE	vs raw / PC1-removed / robust common-mode, paired p < 1e-4
Step-size invariance	max deviation 0.006	delta from 0.01x to 2x mean nodal demand, four placements
Optimal step size	0.5 x mean_d (0.45% of total demand)	Maxwell-Betti asymmetry minimum, 3.11e-04
Directed supply chain	A -> B -> C -> E -> D	flow transit; verdicts identical at all 10 delta levels
Pressure coupling	symmetric to 3.11e-04	reciprocity theorem; pressure can never yield direction
Placement criterion	sigma_min of the district Jacobian	predicts robustness to Jacobian error, not recovery level

2. What the method is

Four stages, each validated against a case whose answer is known by construction before being applied to Graeme.

Stage	What it does	Design decision and why
Flat baseline	All demands constant and pattern-free, duration=0, one steady-state solve.	EPANET's simulation is quasi-static, so a zero-duration solve yields a Jacobian rather than a dynamic response. A longer run would let tanks cycle and reintroduce the drift the design removes.
Perturbation sweep	Each of 113 junctions perturbed individually at a ladder of additive levels.	Additive, not multiplicative: keeps magnitudes comparable across nodes and still probes the many zero-demand junctions a multiplier would leave untouched.
District Jacobian + unmixing	Aggregate node columns to districts; solve $y = J_d d$ by least squares.	Node-level inversion is hopeless (15 sensors, 113 nodes). District aggregation makes it overdetermined (15 x 5). Dependence statistics are then computed on the demand estimate, not on the mixed observation.
Placement by conditioning	Select sensors to maximise sigma_min of J_d .	Unmixing error scales as $1/\sigma_{\min}$, so optimal experiment design is the natural placement objective — better founded than AUROC-against-adjacency or decoupling.

3. Physics validation

3.1 Standard checks

Check	Result	Verdict	What it catches
Mass balance	abs error ~2e-8 m3/s (at output precision)	PASS	Sign and bookkeeping errors. Caught the 0.2 demand multiplier in v1.

Check	Result	Verdict	What it catches
Diagonal dominance	self_is_max 1.000 self/max min 1.000	VACUOUS	Nothing. On a network where pressure moves near-uniformly, self=max is tautological. This check cannot fail and should be removed or redesigned.
Baseline flatness	flow CV 3.7e-07	PASS	Tank cycling that would reintroduce the removed confound.
Linearity (flow)	p05–p95 ratio 0.98–1.05	PASS	How far the linear map extrapolates. Use the percentile band, not max_abs_dev (see 6.4).
Feasibility	100% feasible at every level; min pressure 35.6 -> 18.6 mca	PASS	Survivorship. drop_infeasible never fired, so all levels are computed on the same 113-node set.

3.2 Maxwell-Betti reciprocity, in three tiers

Steady flow linearised about an operating point gives $L \, dH = dd$ with $L = A \, G \, A'$, where A is the node-arc incidence matrix and G is diagonal with positive entries $g = 1/(n \, r \, |Q|^{(n-1)})$. L is a weighted graph Laplacian, hence symmetric by construction, hence L -plus is symmetric. Therefore $dH_j/dd_i = dH_i/dd_j$ exactly: **pressure sensitivity measures coupling strength and has no antisymmetric part to carry orientation**. Direction must come from flow, which is a routing matrix and manifestly asymmetric. This is a theorem, not an empirical regularity.

Tier	What it tests	Result	Verdict
T1	The theorem itself, elementwise over 6328 unordered junction pairs, against the numerical precision floor.	max asym 0.0154; median 0.0031; relative asymmetry 3.11e-04 ; 100% of pairs within floor.	PASS
T2a	Mechanism: is the assembled Laplacian actually symmetric? Independent of the sweep entirely.	$\ L - L'\ = \mathbf{0.000e+00}$ exact; $\ L\text{-plus} - (L\text{-plus})'\ = 1.06e-10$; all conductances > 0 ; one zero eigenvalue (3.2e-15) as expected for an ungrounded Laplacian.	PASS
T2b	Cross-validation: does the analytic L-plus agree with the measured sweep?	corr = -0.5696 . Two defects in the assembly, both mine: the sign ($dH = -L\text{-plus} \, dd$, minus omitted) and the grounding (reservoir arcs dropped, so a floating Laplacian was pseudo-inverted instead of the reduced Dirichlet Laplacian).	INVALID
T3	Falsification: a check valve is non-reciprocal, so inserting one MUST break symmetry.	rel_asym 3.11e-04 both with and without the valve on pipe 96. The valve never closes (baseline flow forward, never reversed by the perturbation), or wntr did not apply the attribute.	NO POWER

Reading. The reciprocity claim rests on T1 plus the exact symmetry of L in T2a, which is sufficient for the theorem. But T2b and T3 are owed: an unfalsifiable check is not evidence, and the cross-validation against closed form is the strongest available statement about the probe itself. Both are cheap to fix and are listed in Section 8.

3.3 The precision floor was mis-estimated, and the correction is informative

The floor was computed as $\sqrt{2} \times 0.005 / \delta = 5.70$, assuming EPANET's 2-decimal report rounding. Measured asymmetry is 370x below that, so the assumption is wrong: wntr reads the binary output at float32, not the rounded report. The true floor is float32 relative precision on a ~35 mca pressure, about 4.2e-6 absolute, giving ~3.4e-03 after division by δ . **Median measured asymmetry is 0.0031**. The measurement sits exactly on the floor — the strongest possible pass, and it explains the shape of the delta curve in Section 4.

4. Step size is a real parameter, now calibrated

Sensitivity is a finite difference, so δ trades cancellation error (small δ) against truncation error (large δ). Reciprocity asymmetry is the ideal selection criterion because it is anchored to a theorem and depends on no downstream metric and no arbitrary reference point.

delta / mean_d	% of total demand	rel. asymmetry	delta x asym	pipes reversed	regime
0.01	0.009	9.32e-03	—	0.0000	precision-limited
0.10	0.089	9.34e-04	0.0046	0.0002	precision-limited
0.25	0.223	3.73e-04	0.0046	0.0003	crossover
0.50	0.446	3.11e-04	0.0077	0.0004	minimum
1.00	0.893	3.41e-04	0.0169	0.0009	curvature-limited
2.00	1.786	6.02e-04	0.0600	0.0023	curvature-limited
5.00	4.464	1.20e-03	0.3014	0.0058	curvature-limited

delta / mean_d	% of total demand	rel. asymmetry	delta x [asym]	pipes reversed	regime
100.50	89.73	1.15e-02	—	0.0570	symmetric = FALSE

The **delta x [asym]** column is decisive. Cancellation asymmetry falls as $1/\text{delta}$, so a constant product means precision-limited; it is constant at 0.0046 for 0.1x and 0.25x, then grows. The crossover sits between 0.25x and 0.5x and the minimum at 0.5x. **Adopted: $\text{delta} = 0.5 \times \text{mean_d}$.**

4.1 Large perturbations reveal a validity boundary, not hidden coupling

At 100.5x mean_d (89.7% of total system demand injected at one node) District A's influence map lights up where it was previously dark. This is not concealed coupling. At that level 5.7% of pipes reverse flow, minimum pressure falls 48%, and reciprocity fails its own threshold. The routing matrix itself has changed — a second operating point, not a derivative. A noise-floor explanation is ruled out: apparent sensitivity from solver noise would scale as $1/\text{delta}$ and therefore *fall* with delta, whereas the observed values rise.

This is the quantitative version of v1's hand-waved limit. Reportable as: reciprocity holds to $3\text{e-}04$ up to 1x mean nodal demand and fails past ~90% of total system demand, where 5.7% of pipes reverse.

5. Recovery results, pinned set

40 worlds, variant = baseline, n_months = 30, explicitly pinned by sim_hash. Truth is the pairwise correlation of district-aggregated node demands; the estimate is the same statistic computed on the recovered demand. Rank correlation is Spearman over the 10 district pairs; MAE is on the correlations themselves.

Placement	sigma_min	raw	PC1-removed	robust CM	unmixed	gain	paired p
variance	0.281	0.007	0.013	0.035	0.939	+0.767	<1e-4
decoupled	0.040	0.304	-0.009	0.039	0.847	+0.464	<1e-4
manual	0.098	0.187	0.255	0.170	0.756	+0.338	<1e-4
boundary	0.167	0.220	0.389	0.194	0.709	+0.223	0.0057

MAE separates the methods more sharply than rank correlation: unmixed 0.052–0.154 against 0.70–0.82 for every model-free method. The free methods preserve some rank ordering but get the *level* of the correlations badly wrong, which matters because the downstream statistics (excess similarity, attribution share) need calibrated values, not just ranks.

5.1 sigma_min predicts robustness, not level

Ordering by sigma_min does not order recovery (decoupled at 0.040 beats boundary at 0.167). What sigma_min does predict is **degradation under Jacobian error**, since unmixing error scales as $1/\text{sigma_min}$. Three independent stress tests now give the same ranking:

- **Wrong node weights.** variance is flat (slope +0.007 in rank corr vs weight error; 0.955 at the oracle, 0.955 with uniform weights, 0.945–0.970 across lognormal sigma up to 1.0). boundary, manual and decoupled all degrade.
- **Spatially coherent weight error.** A BFS-concentrated drift at weight error 0.097 does more damage than isotropic error at 0.303 — variance -0.050, boundary -0.107, decoupled -0.219, manual -0.298. Cosine distance is therefore *not* a sufficient statistic for misspecification harm.
- **Nonlinear Jacobian perturbation.** Across the delta ladder, decoupled (sigma_min 0.040) is the only placement that degrades at large delta: 0.847 -> 0.734, MAE 0.095 -> 0.166, while the other three improve slightly.

Consequence for deployment. variance is purely local, needs no Jacobian and no coordination to compute, and is the most robust to every form of model error tested. That is the right recommendation for a federated setting, and it now rests on robustness rather than on a marginal conditioning advantage.

5.2 Step-size invariance

Recovery is flat across 0.01x to 2x mean_d: maximum deviation from the 0.5x reference is 0.006 (boundary), 0.003 (decoupled), 0.006 (manual), 0.002 (variance) — all well inside the ~0.02 noise floor of a Spearman over 10 pairs. The choice of delta was never load-bearing for the headline.

5.3 The honest sample size

The 40 worlds are drawn from only **4 distinct consumption maps**, and every world shares the same network, so Jd and the hydraulic similarity matrix are literally constant across them. The n=40 paired p-values above treat correlated draws as independent and should not be quoted. Per-map mean gains are +0.704, +0.796, +0.515, +0.588 — all four positive, exact sign test **p = 0.0625**, which does not clear 0.05. **Cluster count is currently the binding constraint on this result**, and it

is the main thing the `jval_*` suite fixes.

One negative cell, worth diagnosing rather than averaging away: map LR_LC_LI_LR_LR with the boundary placement gives a gain of -0.359, and it worsened when the sample doubled (-0.279 at $n=20$), so it is systematic, not noise. That map has three residential districts, so several true pairwise correlations should be near-tied and Spearman over ties is unstable by construction. Check whether MAE also degrades there; if not, the metric is at fault, not the method.

6. Corrections to v1

Recorded as wrong rather than quietly dropped, in the same spirit as v1's dead-end log.

#	v1 claim	Status	What is true
6.1	Recovery 0.955 +/- 0.040; boundary 0.832, decoupled 0.892, manual 0.801, variance 0.955.	SUPERSEDED	Those figures came from a world set selected by <code>.head(20)</code> on a pool that grew from 43 to 45 to 60 as new studies ran, silently swapping 4 of the 20. They cannot be reconstructed. Pinned values are 0.709 / 0.847 / 0.756 / 0.939. Delta was <i>not</i> the cause: the legacy Jacobian differs from the 0.5x one by at most 0.024.
6.2	Directed supply chain A -> B -> C -> D <-> E.	WRONG	The order is A -> B -> C -> E -> D . All 10 pairs are directed at all 10 delta levels with byte-identical verdicts; D/E is not mutual. But it is the <i>weakest</i> edge: its asymmetry is the only one that grows with delta (0.0186 -> 0.0895, ~5x), the delta->0 limit is ~0.0185 against a tolerance of ~0.012, so it clears by 1.5x where every other pair clears by 25–80x. Report D/E as weakly directed.
6.3	Mass balance 4.85e-06 relative error.	ARTIFACT	<code>rel_error</code> = <code>abs_error</code> / delta, and the numerator is fixed by output precision at ~2e-8 m3/s. The relative figure falls monotonically with delta (5.1e-04 to 0) and says nothing about accuracy. Quote the absolute closure error, which is delta-independent.
6.4	Linearity bounded, <code>max_abs_dev_from_1</code> reported.	ARTIFACT	That statistic is non-monotone in delta (11.5, 19.8, 847.8 , 5.9, 15.3, ...), with its spike where nonlinearity is negligible. Genuine curvature must be monotone. It is near-zero-flow pipes with exploding ratios. Use the p05–p95 band, or threshold by <code>min_norm_pct</code> .
6.5	78% of observed similarity is hydraulic mixing.	ILL-POSED	A ratio of correlation coefficients does not decompose variance. Four pairs exceed 100% (up to 135%), which is the diagnostic that the estimand is not a share. It is also unstable across choices made after computing it: 89.3% uniform, 79.3–86.6% across preprocessing, 78.0% demand-weighted. Needs a covariance-scale definition or a partial-correlation framing.
6.6	Sector structure retained after unmixing, 93–100% of the true gap, $p = 5.0e-06$.	UNVERIFIED	Not present in any saved notebook output. The only sector test in the artifacts gives $p = 0.154$ with the opposite sign . Either the cell is lost and must be re-run and committed, or the claim is withdrawn. The same applies to 'Jacobian prediction vs simulated independent demands, agree to 0.01'.
6.7	<code>sigma_min</code> predicts recovery level and noise decay.	HALF	It predicts noise decay and robustness to Jacobian error. It does not order recovery level (manual 0.098 -> 0.756; decoupled 0.040 -> 0.847).
6.8	Income regime untestable because all maps are all-low income.	DEEPER	True, but the deeper problem is that the measurand is scale-blind. Pearson correlation is invariant to amplitude, and income is a level statement, so varying income would not move the ground truth at all . Income needs a second, level-sensitive measurand (see 7.2).

6.9 Process finding: pin the world set

The v1 numbers are not wrong so much as *irreproducible*. Every result quoted so far used `.head(20)` on a pool that changed at least twice, so "20 worlds" has meant different sets of 20 at different points. Given that a stale-estimate bug already cost this project a headline once, the world set is now a hardcoded list of `sim_hashes` and the filter is explicit (variant, `n_months`). Cheap insurance, and it should be standard for every table that goes into the thesis.

Two secondary hygiene items found the same way: notebook variable collisions (a scratch assignment to B destroyed the ladder results frame, and D shadows the supply-dependency matrix), and `unmix_demands(rcond=1e-3)` never forwards `rcond` to `pinv` — the parameter is dead and no regularisation runs. Harmless at cond 5–15, but not what the signature claims, and it will matter for the badly-conditioned arms of `jval_*`.

7. The `jval_*` suite: what it must establish

Status. Worlds generated at `n_months = 48`. They are excluded from the Section 5 analysis by the `n_months == 30` filter, which is correct — they need windowed pre/transition/post analysis, not the whole-series statistic. No `jval` result exists yet.

7.1 The three questions

Q	Question	Why it is open	Decides
Q1	Does the recovery result hold with a defensible number of independent clusters?	4 consumption maps, all worlds at sim_seed 42, one network. Sign test $p = 0.0625$.	Whether Section 5 is reportable at all.
Q2	Does the Jacobian survive a regime change, and does a client need to recommission it?	The Jacobian is linear about a flat baseline while the worlds drift. v1 measured recovery over the whole series, mixing pre- and post-drift.	The deployment claim: commission once, or recalibrate.
Q3	Is the level (income) axis recoverable at all?	Never tested. Per-client MinMax in the FL path destroys level; the unmixing path is a linear inverse and preserves it.	Whether unmixing sees regime structure the FL artifacts cannot.

7.2 Two measurands, deliberately orthogonal

	M1 — structure	M2 — level
Statistic	rank corr and MAE of pairwise correlations; gap retention $G_{\text{hat}} / G_{\text{true}}$ where $G = \text{mean } \rho(\text{same-state pairs}) - \text{mean } \rho(\text{different-state pairs})$.	$r[d] = \log(\text{std}(d_{\text{hat}}[d]) / \text{std}(d_{\text{true}}[d]))$. Report std(r) across districts and the Spearman rank of districts by amplitude.
Sensitivity	Scale-blind, shape-sensitive. Reads land use .	Shape-blind, scale-sensitive. Reads income .
Caveat	Cannot see income by construction. On the income arm M1 is a null control : if it moves across the drift, that is method error.	Jd carries a global scale factor, so mean(r) is unidentifiable and diagnostic only. Only relative amplitude and rank are reportable.

7.3 Horizon and windows, verified empirically

Established from gt_drift_schedule.csv across 45 legacy worlds, with no new simulation:

- **Onset equals warmup_months exactly.** First switch is month 4 at warmup 4, month 12 at warmup 12.
- **The front is fully predictable.** Seed converts alone, then max_neighbors_per_month = 2 per advance, advancing with growth_chance = 0.8. active_months = $1 + \text{ceil}((n-1)/2)$, matched exactly for all five districts: A 7, C 10, D 13, B 15, E 16.
- **Span is independent of warmup_months** (District D: span 18 at both warmup 4 and 12), so warmup is a pure offset.
- **The schedule is deterministic at fixed sim_seed** — identical onset, completion and stall pattern across all 8 worlds per district, regardless of land use or map. The conversion path therefore contributes *zero variance* to every existing result. sim_seed is the only axis that varies it.
- **Horizon is a pure suffix.** A 36-month and a 48-month world are bitwise identical over the shared 36 months, drift onset included. So n_months is not gridded: 48 subsumes every shorter horizon and short horizons become window selections made in analysis.

Window	Months	Steps	Rationale
pre	0–5	4 320	Onset at 6, so uncontaminated with zero margin needed.
transition	6–35	—	The front advancing. Not waste — see 7.5.
post	36–41	4 320	$36 = 0 \bmod 12$, so pre and post cover the same calendar months and the seasonal cycle cancels rather than merely matching. Equal lengths, so the two correlation estimators have identical variance.
spare	42–47	—	Margin. Worst case is District E: 16 advances at $p=0.8$ gives a p99 span of ~26 months, completion ~32 including the 42-day ramp.

7.4 Arms

Study	Design	Worlds	Purpose
jval_beta	beta in {0, 0.35, 1.0}, one drift origin, one seed.	3	Mechanism. beta dials the land-use level effect. At beta=0 the drift is volume-neutral so the operating point barely moves and the Jacobian should hold; at beta=1 the excursion is largest and $h \sim Q^{1.85}$ curvature should bite. Monotone degradation identifies the operating point as the cause. Also the feasibility canary — run first, 3 worlds is cheap to lose.
jval_coupling	baseline vs isolated, two drift origins (A the trunk, E the parallel end), one seed.	4 (2 cached)	Falsification. With close_fraction 1.0 there is no mixing to undo, so unmixing must show near-zero gain . If the gain persists, attributing it to hydraulic mixing is wrong. Strongest available test that +0.767 means what we say.
jval_income	LR x4 + HR, land use held residential; 2 drift origins x 3 seeds.	6	Q3 / M2. Pure level contrast: low 9.25 to high 33.00 m3/month/unit, ~3.6x amplitude. M1 is the null control here.

Study	Design	Worlds	Purpose
jval_s1_oddone	LR x4 + LI; one client drifts to industrial; 5 origins x 3 seeds.	15	Q1 / Q2. Carries two properties nothing else does — see 7.5.
jval_s2_split	LR x3 + LI x2; one client crosses; 5 crossings x 3 seeds.	15	Better-conditioned truth than S1: 4 same-state and 6 different-state pairs, so the gap statistic G is estimated from both cells rather than one.
jval_s1_equalised	S1 with max_neighbors_per_month tuned per district (A 2, C 3, D 4, B 4, E 5) for 8 active months each.	5 (optional)	De-confounds transition duration from district size. Only if a per-district difference actually appears.

7.5 The two properties that make S1 the strongest arm

An invariance null inside every world. When District D drifts, $\rho(A,B)$, $\rho(A,C)$ and $\rho(B,C)$ are unchanged by construction. Any movement in the *estimated* value is pure method error, measured with no ground-truth ambiguity and no need for a separate control world.

A signed prediction. $\rho(D,E)$ must rise; $\rho(D, \{A,B,C\})$ must fall. Recovering the signed *change* induced by a known intervention is a causal validation, not a correlational one. This is a stronger claim than anything in Section 5.

And the transition window is a prediction, not waste. Mid-conversion a district is internally heterogeneous, which is exactly the spatially coherent node_weights error that was the only thing that hurt recovery in the misspecification test. Predicted signature: recovery dips during transition and returns afterwards. Error tracking a known physical cause and then recovering is much stronger evidence than a flat robustness number.

7.6 Analysis arms, zero world cost

Topology does not change under drift, only demands. So a Jacobian changes only through the operating point and through within-district composition, and those are separable at the cost of a few hundred solves.

Arm	Jd built from	Isolates
J_flat	the .inp base demands	the Section 5 baseline
J_pre	flat probe at pre-drift mean demand	commissioning-time J
J_post	flat probe at post-drift mean demand	recommissioned J
J_pre_wpost	J_pre structure, post-drift node_weights	composition alone
J_perm	district columns of Jd shuffled	falsification: recovery must collapse

If J_pre holds in the post window, the deployment claim becomes **commission once, no recalibration after a regime change** — a materially better story than "we re-measured it".

7.7 Per-world assertions, to run before including any world

Three lines that would have caught the resolution_probe truncation the day it happened (those worlds converted 5 of 24 nodes with the 42-day ramp clamped to 29 days, and the finer-resolution conclusion was drawn on them):

- `len(sch) == n_nodes[tgt]` — the front completed.
- `sch.drift_month.max() + 1.4 < 36` — it completed before the post window opens.
- `sch.drift_month.min() >= 6` — the pre window is uncontaminated.

7.8 Pre-registered predictions

Written down before the numbers exist, precisely because a post-hoc-selection bug has already bitten once.

#	Prediction	If it fails
P1	Pre-window MAE for variance reproduces Section 5 (~ 0.05).	Something is wrong with the windowing, not with the method.
P2	Transition-window recovery is the worst of the three windows.	Composition error is not the dominant failure mode; look at the operating point instead.
P3	Invariance: $ \delta \rho_{\text{hat}} $ for non-drifter pairs stays inside the ~ 0.02 noise floor.	The estimator is not isolating districts; a serious problem.
P4	Signed change for drifter pairs has the correct sign in $\geq 90\%$ of cases.	The causal claim is withdrawn; recovery stays correlational.

#	Prediction	If it fails
P5	Isolated worlds show near-zero unmixing gain.	The +0.767 gain is not attributable to hydraulic mixing. This one is load-bearing.
P6	Degradation is monotone in beta.	The operating point is not the mechanism; look for composition or conditioning.
P7	M2 ranks districts by income correctly.	Level is not recoverable and Q3 answers negatively — still worth reporting.

8. Next steps, in dependency order

#	Step	Why now	Cost
1	Fix and re-run T2b (add the minus sign, ground the Laplacian at the reservoir) and T3 (pick a near-zero-baseline-flow loop pipe, or verify the CV reaches the written .inp).	The reciprocity claim currently has no demonstrated falsifier, and the closed-form cross-validation is the strongest statement available about the probe itself.	Low
2	Recover the missing fifth consumption map, or widen the filter to $n_months \geq 30$ and window the longer worlds down.	4 clusters gives a sign test at $p = 0.0625$. This is the only thing between Section 5 and a reportable result.	Low
3	Diagnose the LR_LC_LI_LR_LR / boundary negative cell: inspect whether the true correlation matrix is near-tied there.	The one systematic negative in 16 cells. Likely a metric artifact, but unverified.	Low
4	Run jval_beta (3 worlds) as the feasibility canary, then jval_coupling, then the rest.	beta=1.0 with two industrial districts is the highest infeasibility risk in the suite. Better to find that in 3 worlds than in 41.	Low
5	Forward rcond in unmix_demands before trusting any badly-conditioned arm.	The parameter is currently dead. jval's decoupled and uniform-weight arms will push cond up.	Trivial
6	Bridge to the FL pipeline: unmix as a preprocessing step before fl_preprocessing.	Unmixed demand is locally computable by each client from its own sensors plus one Jacobian column. Tests whether federated artifacts recover structure that unmixing makes visible.	Medium
7	Shape-based regime statistic alongside correlation.	Sectors differ in profile shape; correlation captures co-movement. A normalised-weekly-profile distance may be the sharper instrument, and it survives per-client MinMax where level does not.	Medium
8	Multi-source / tank network.	Mixing becomes state-dependent and a single static Jacobian may not suffice. The main external-validity limit.	High

9. Limits to state plainly

- **One network, one operating point, static.** The Jacobian is valid near the baseline it was measured at. Nonlinearity is now bounded rather than assumed (Section 4), but it is not zero.
- **jval_* does not address external validity.** The "n that matters" differs by claim: for the estimator claim, clusters are consumption maps and seeds, and new worlds help; for the Jacobian-validity claim, clusters are *networks*, and 41 new worlds move that from $n=1$ to $n=1$. Only a second network would.
- **Graeme has a single reservoir and no tanks.** Both the pressure-uniformity finding and the sufficiency of one static Jacobian may not transfer.
- **Demand-driven model.** Under PDD the Jacobian gains a demand-feedback term. Reciprocity survives if the PDD curve is common across nodes, but the conductances change.
- **Check valves, PRVs and active pumps break reciprocity legitimately.** Graeme has none active. On a network that does, the asymmetry would be informative rather than an error — which is also why T3 matters.
- **Transition duration is a deterministic function of district size,** and size is collinear with hydraulic position. "Which district drifts" bundles size, position and conversion rate, and five districts cannot de-confound them. jval_s1_equalised is the partial remedy.
- **node_weights assumes district demand moves proportionally to node size.** BFS-concentrated drift violates this, and it is the one form of misspecification that measurably hurts. variance is nearly immune; the other placements are

not.

10. Artifacts

File	Contents
sensitivity_probe.py	Flat baseline, perturbation sweep, physics checks, influence maps, supply dependency, similarity attribution, unmixing, placement selectors (decoupled, D-optimal), ipywidgets explorer.
influence_probe.ipynb	The interventional study end to end: probe and maps, attribution, placement, unmixing, noise robustness, delta ladder, pinned reconciliation, Maxwell-Betti tiers.
placement_poc.py / dependence_and_placement.ipynb	Observational side: placement strategies, dependence battery, horizon sensitivity, directional diagnostics — the work the probe explains.
experiments_jval.yml	The six jval_* studies: frozen world block (oracle matched to d0_replication so the worlds stay FL-reusable), n_months 48, warmup_months 6, worlds-only run block.

Every number above is anchored to a saved notebook output on an explicitly pinned world set, and every method was validated against a construction with a known answer before being applied to Graeme. Where a v1 claim did not survive that discipline, Section 6 records it as wrong rather than restating it.